Embracing not avoiding generative AI to enhance scientific writing education

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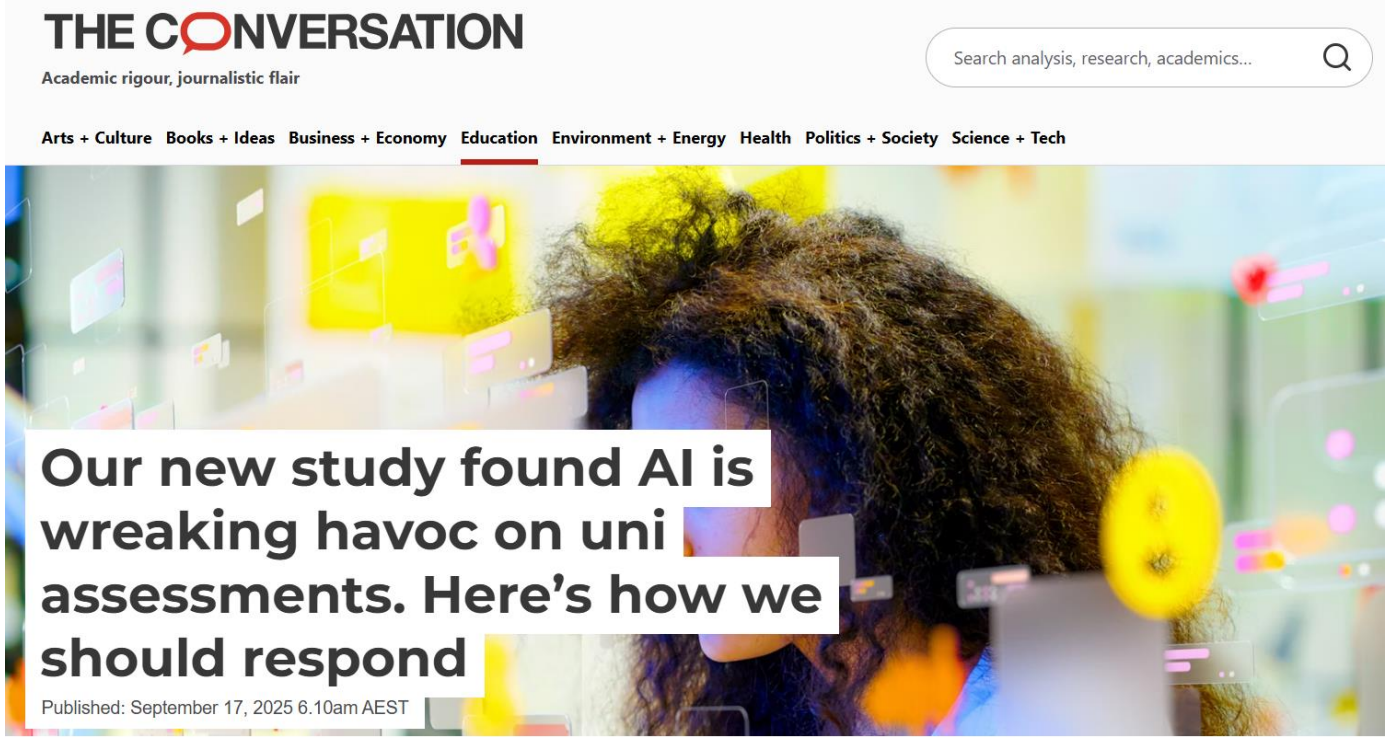
GenAI – A wicked problem

Significant impact on university assessment and assurance of learning outcomes

- A ‘wicked problem’ is one without a neat solution – often necessitates trade-offs in response
- GenAI is here to stay, and students are using it
- Expectation to redesign assessment while technology is still changing
- Accept AI in assessment is not a simple solvable problem



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The Challenge – A changing landscape

Modernising assessment in Cell and Molecular biology

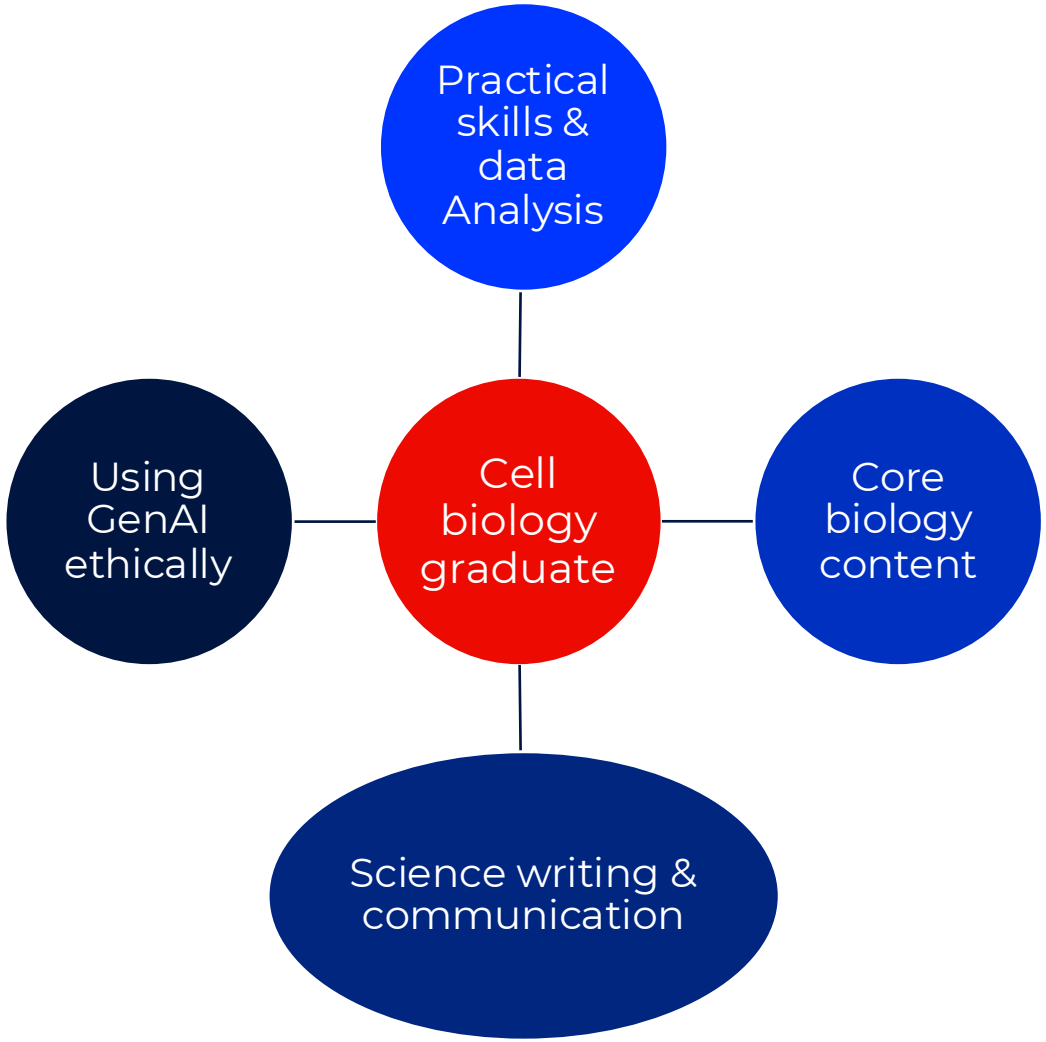
- Computational skills including using AI are now an essential graduate skill
- Must still provide our core education in biology, practical skills and communication

SO….

- Design assessments that both assure learning and model ethical engagement with emerging technologies
- GenAI in assessment is largely undetectable
- The technology is changing

The Challenge – A changing landscape

Modernizing assessment in Cell and Molecular biology



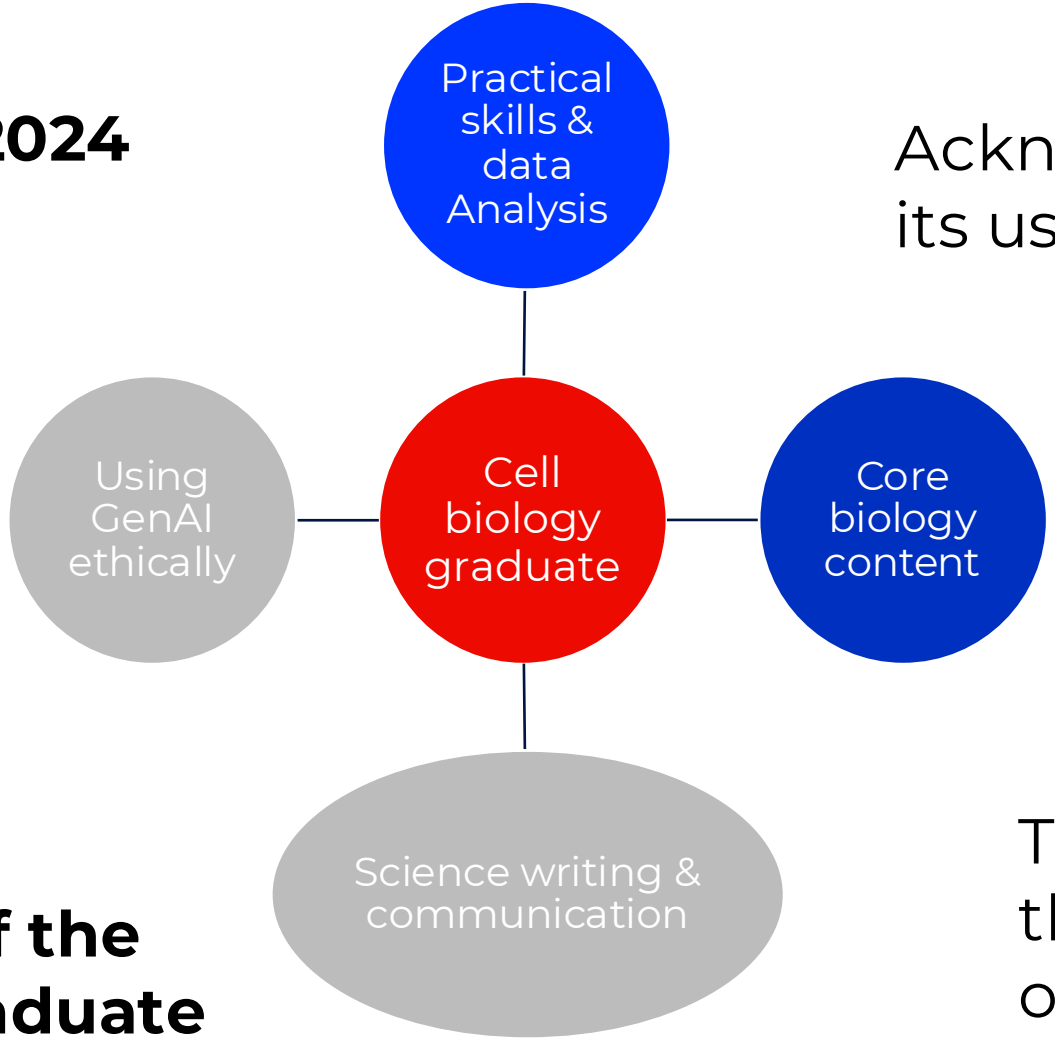
The skills of the modern graduate

The Challenge – A changing landscape

Modernising assessment in Cell and Molecular biology

Biol340 in <2024

Acknowledges its use



Creates a more equitable learning environment

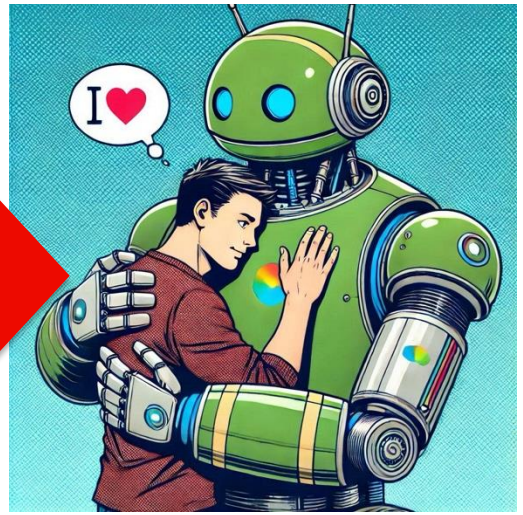
The skills of the modern graduate

Transparency into student thinking to assure learning outcomes

The Plan – Embrace not avoid Gen AI

Redesigning Assessment

Adapt our assessments to include generative AI as a core component



- Two new assessments framed around scientific writing, review and reflection were introduced
- Linked to practical project and data generated in wet labs
- ‘Dry’ practicals were added to
  - support data analysis
  - scaffold scientific writing and review
  - improve GenAI literacy

The Action – Scientific Writing, Review and Reflection

1. Peer Feedback on Scientific Writing

- Students worked in pairs to develop a piece of formal scientific writing
- Reports were peer marked following established rubric and written feedback provided
- Student sees a variety of reports and can reflect on writing/communication
- Report then served as draft for reflection assignment

Assessment 1, worth 15%, was group work and peer assessed

The Action – Scientific Writing, Review and Reflection

2. GenAI Feedback on Scientific Writing

Activity 1: Prompt design and output exploration

- test the prompts on a piece of writing

Activity 2: Compare and evaluate

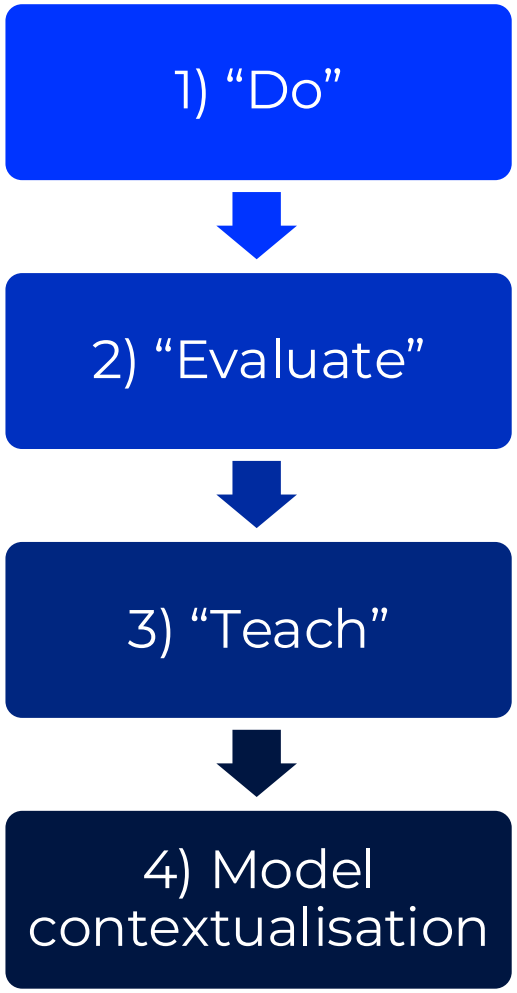
- share prompts and output and consider which was better and why

Activity 3: Ethics in action

- discussion around when is it ok to use GenAI and when is it not – 'how to maintain ownership'

Activity 4: Collaborative rewrite

- Revise a paragraph with AI and consider what was changed and why



The Action – Scientific Writing, Review and Reflection

2. GenAI Feedback on Scientific Writing

By guiding students through GenAI use we hoped to:

- Teach prompt engineering fundamentals
- Minimise unethical AI use (i.e. direct copy-pasting without attribution)
- Improve confidence in scientific writing & editing

"I liked how we were encouraged to use AI to aid our learning rather than discouraged from it"

Part 2: Revising Your Report Using AI

Instructions: Experiment with the following three levels of prompts while revising your report. Document your prompts, AI outputs, and how you applied the feedback.

Prompting Log

2.0. Contextualise the model

- **Prompt used:**
  (e.g., "You are an expert in scientific writing and an editor of a top ranked journal")

2.1. "Do for me" Prompt

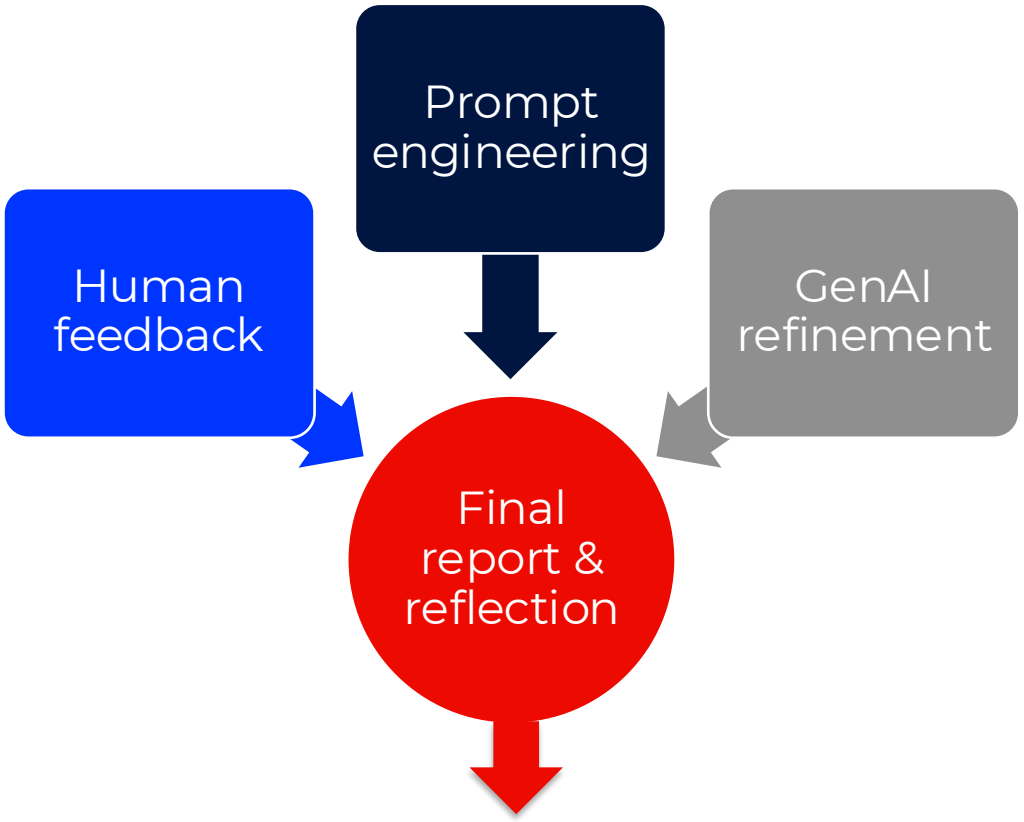
- **Prompt used:**
  (e.g., "Rewrite my discussion section to make it clearer.")

- **AI Output:**
  (Paste the relevant output here)

The Action – Scientific Writing, Review and Reflection

3. Reflection on Feedback

- Reflected on human vs. AI feedback and the use of AI tools for scientific writing
  - *How did Copilot feedback compare with the feedback from your peer?*
  - *Which type of Copilot prompt helped most? Why?*
  - *Did using Copilot change how you approach writing or editing?*
  - *What are the benefits and risks of using GenAI like Copilot in science writing?*



Compared their drafts pre- and post-refinement and discussed how AI changed their work

The Action – Scientific Writing, Review and Reflection

The Final Submission

- Revised report with tracked changes
- Completed proforma with
  - feedback summary
  - GenAI prompts
  - REFLECTION
- Transcripts of Copilot interactions

PART 3: REFLECTION

Instructions: On the following page write a **~600-word reflection** addressing the following questions:

3.1: **Comparison of Feedback:** How did the quality of feedback from AI compare to that of human feedback? (Max 150 words)

3.2: **Effectiveness of Prompting Levels:** Which level of prompting ("do for me," "evaluate," or "teach me") was most effective for improving your report? Why? (Max 150 words)

3.3: **Impact on Your Writing Approach:** Did using AI alter how you approach scientific writing or revision? How? (Max 150 words)

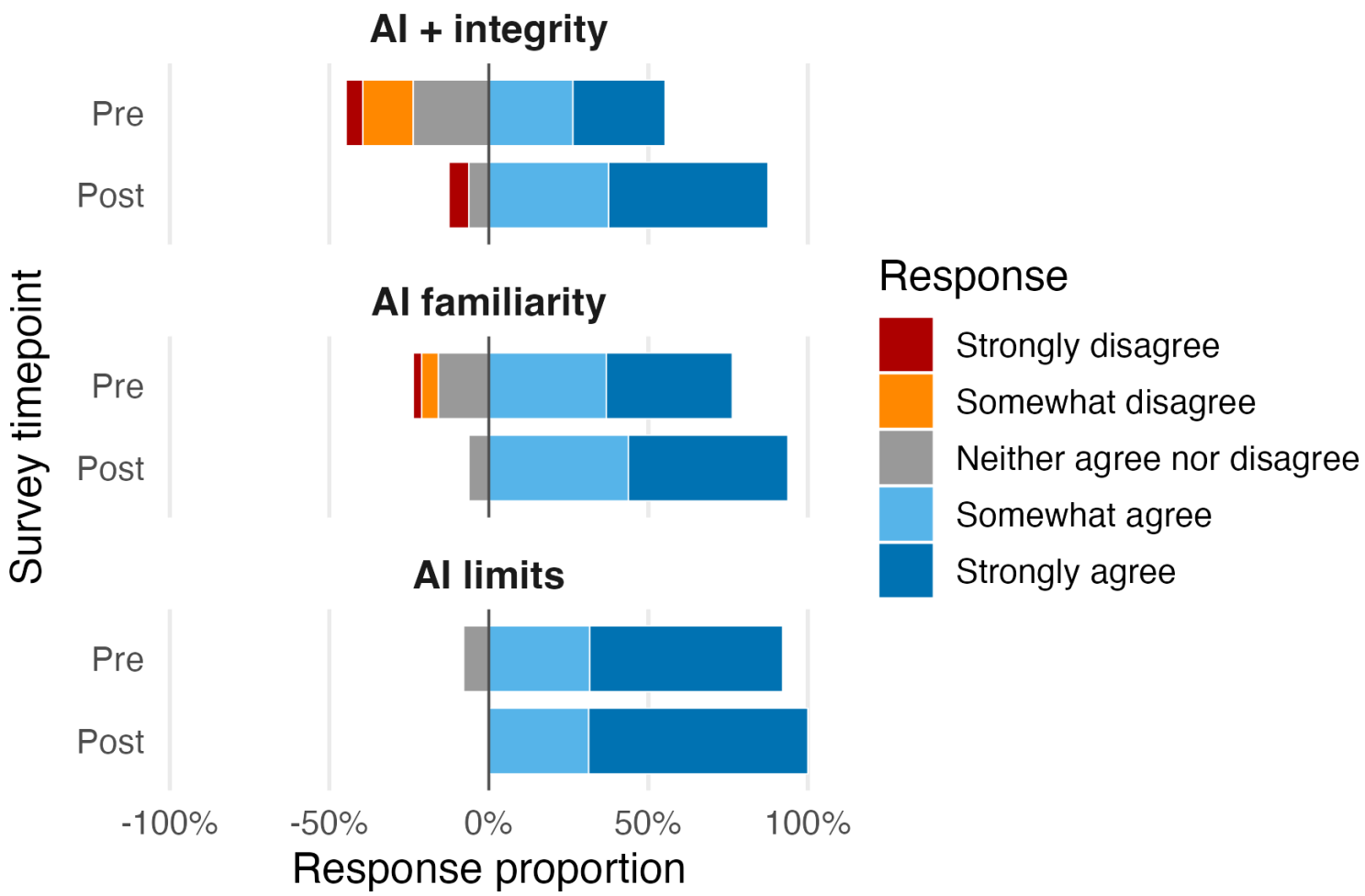
3.4: Limitations and Advantages of AI Tools, ethical AI use: What are the limitations and advantages of using AI tools for scientific writing? What would constitute inappropriate use of AI in this task? Why? (Max 150 words)

Assessment 2, worth 15%, was individual and marked by academics

The Results – Student survey data

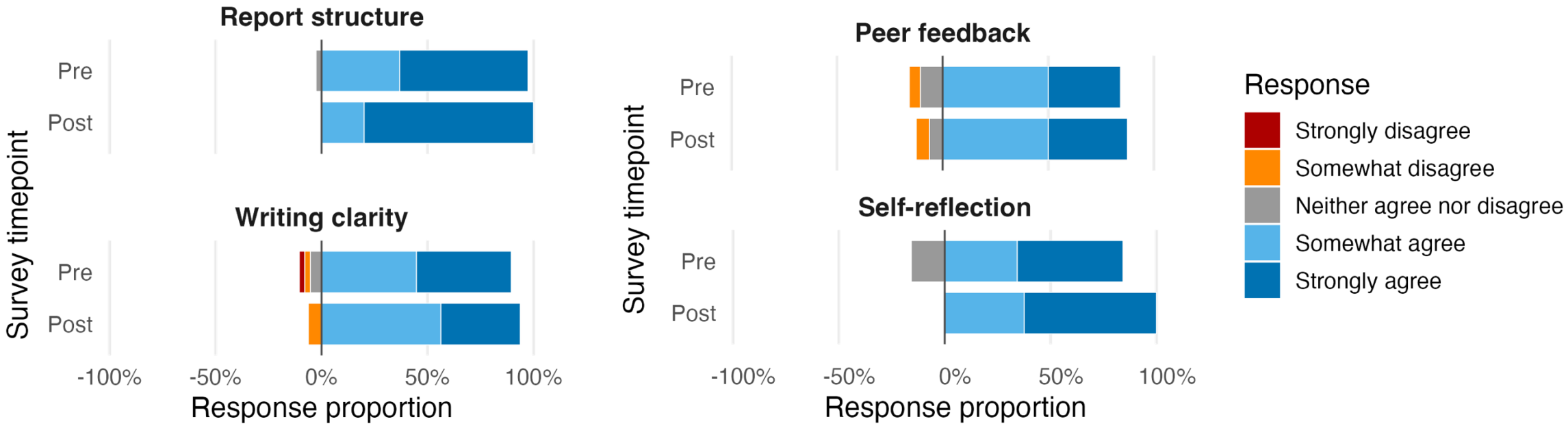
Biggest ‘confidence’ gains in AI familiarity and knowledge

- Students were surveyed pre (44) and post (20) workshops
- Series of statements on their confidence and familiarity with
  - GenAI awareness and use
  - Scientific writing
  - Feedback and reflection



The Results – Student survey data

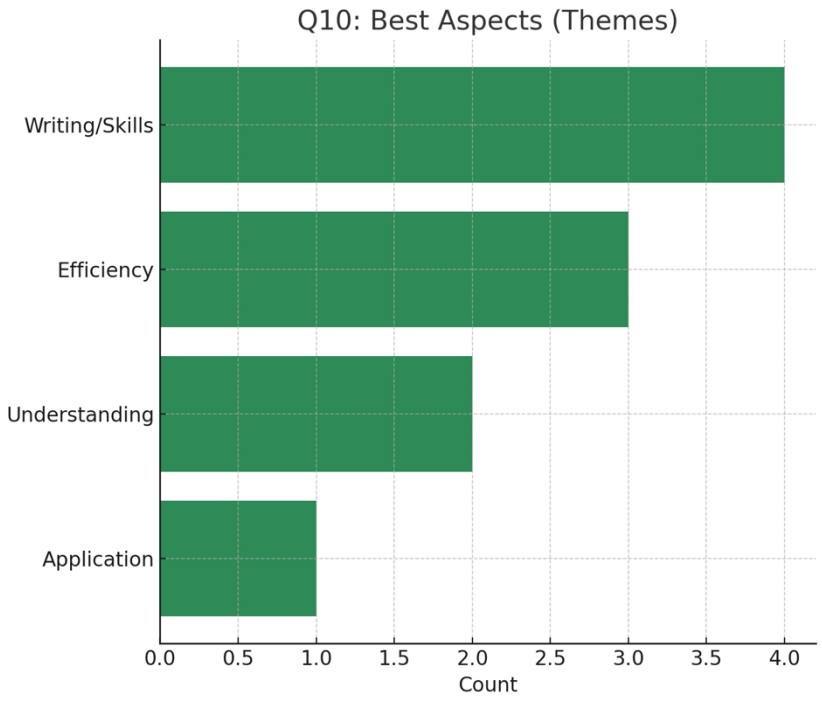
Students felt confident going into the writing task



- Not much difference in pre-post confidence responses BUT open-ended questions and reflection submissions tell a different story

The Reflection – GenAI and Scientific Writing

Writing Improvement



The better I got at prompting, the better the AI became at helping me improve

AI helped me realise I was being repetitive — I rewrote the discussion to focus more on results interpretation

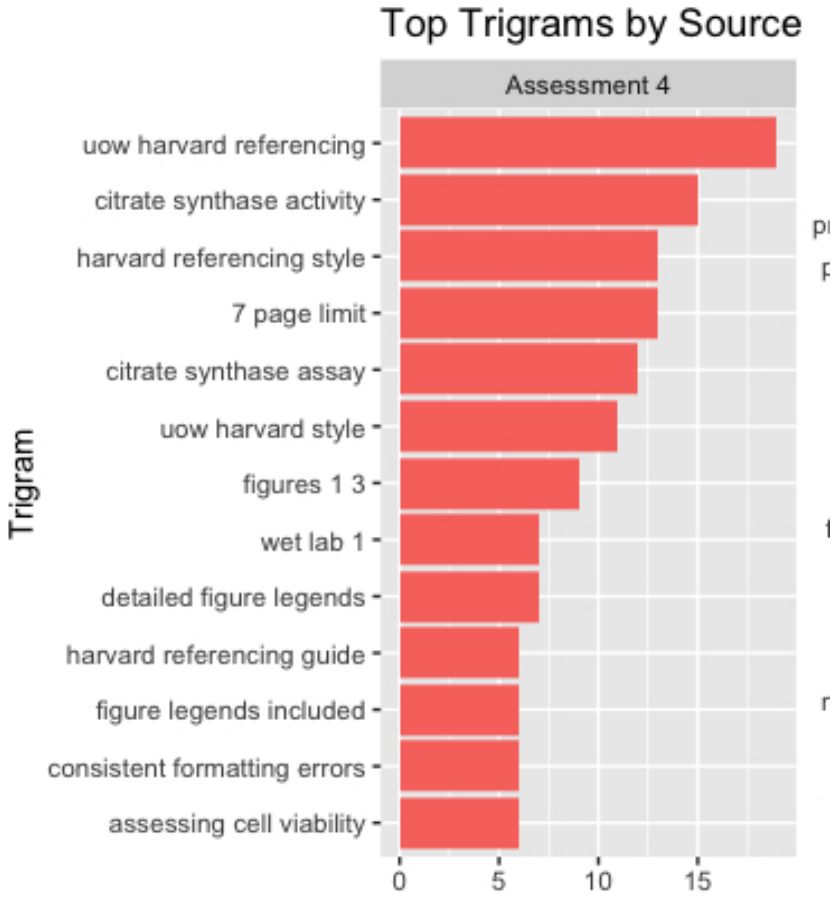
I'm more confident in writing scientific paragraphs after seeing how to link ideas better

"teach me" actually helped me understand why certain choices worked better in scientific writing. I could even show the AI my marking rubric so the feedback aligned directly with what I was being assessed on.

The Reflection – GenAI versus Human Feedback

The consensus was….

- AI feedback was clearer, more actionable for writing structure and tone



AI gave me feedback on how to improve flow and grammar — it was more helpful than the vague 'good job' from my peer.

The peer knew what we were meant to do — AI sometimes gave feedback that didn't match the report format."

I liked that the AI could pinpoint sentence structure issues and suggest alternatives on the spot.

The Reflection – Learnings and takeaways

Embrace not avoid

- Curriculum restructure requires careful consideration of the skills needed by biology graduates
- By encouraging AI use we are equipping students with skills to use this technology more effectively
- Our reflection assessment actively asked for output from GenAI – which discouraged cheating & plagiarism
- Through our approach we were able to reintroduce scientific writing education in an innovative way
- Students seemed to gain from and even enjoy the experience
- Careful consideration of survey questions and approach to data collection in further iterations – reflections are a rich source of information

Questions?

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